SURV/SURVMETH 720 –
Total Survey Error

Weighting
(+ Analytic Error and Imputation)

Class #6

October 6 2025

I. Weighting

- **Weighting** refers to statistical adjustments that are made to survey data after they have been collected whereby each observed element is assigned a weight adjustment factor
- Two main reasons:
  - to correct for unequal probabilities of selection during sampling
  - to help compensate for frame deficiencies and nonresponse (our focus)

Weighting Methods

A. Nonresponse weighting

- Process of weighting up respondents to account for nonrespondents in sample
- Requires unit-level information on characteristics of both respondents and non-respondents

(weighting for non-coverage ?)

B. Post-stratification and Calibration

- Can improve accuracy of estimates
- Does not require unit level information on characteristics of non-respondents/non-covered units

Weighting Methods

A. Nonresponse (or noncoverage) weighting
- **Process of weighting up respondents to account for nonrespondents in sample**
- Requires unit-level information on characteristics of both respondents and non-respondents

B. Post-stratification and Calibration
- Does not require unit level information on characteristics of non-respondents/non-covered units
- Process of adjusting sample to known control totals
  - control totals from some reliable external source
- Can address sample frame deficiencies in addition to nonresponse

A. Nonresponse weighting

1. Adjustment cells

2. Propensity weighting

Nonresponse weighting

- $n = 100$ persons
- Respondents = 50 persons

  - What would be the ‘simplest’ way to account for non-respondents?
  - Will this make a difference to an estimate (say, a proportion) ?
  - What assumption are we making?

Adjustment cell-based weighting

- Classify respondents and nonrespondents into adjustment cells (i.e., "weighting classes”) based on unit characteristics
- Compute response rate within each cell, and use the reciprocal as the weight factor

Geographic Region	Sample Size	Number of Respondents	Response Rate	Nonresponse Weight
Northeast	100	10		
West	75	25		
Midwest	60	30		
South	60	40		

- Idea is that, within subgroups, expected values of respondents = nonrespondents

- Grace, Weishan: how do we determine whether data is missing at random or not ?

"MAR is more general and more realistic than MCAR. Modern missing data methods generally start from the MAR assumption." (p. 7)

"Several tests have been proposed to test MCAR versus MAR. These tests are not widely used, and their practical value is unclear. (p.31). It is not possible to test MAR versus MNAR…" (p. 31)

Flexible Imputation of Missing Data, Stef van Buuren, 2012

NMAR test: https://stefvanbuuren.name/fimd/sec-nonignorable.html

"we want to form classes so that we can claim that we have MAR, i.e., given the *x* 's that define classes, all units have the same response probability.**"**

Practical tools…, Valliant et al.

- Xiaoqing: "How realistic is the assumption that respondents and non-respondents are comparable when using sample-based weighting ?"

- What if adjustment cells have very few cases?
  - Kreuter, F., Lemay, M., & Casas-Cordero, C. (2007, July). Using proxy measures of survey outcomes in post-survey adjustments: Examples from the European Social Survey (ESS). In *Proc. Surv. Res. Meth. Sect. Am. Statist. Ass* (pp. 3142-3149).

In weighting class adjustments only a small set of categorical variables is typically used to ensure cell sizes are sufficient for a stable adjustment. A larger list of these variables as well as continuous measures can be incorporated into a model to predict propensity scores (Little and Rubin, 2002).

A. Nonresponse weighting

1. Adjustment cells

2. Propensity weighting

2. Propensity Score weighting

- Basic procedure:
  - Define response indicator (*r* =1 if sample unit responds, =0 if not)
  - Fit a regression model with *r* as outcome variable and unit characteristics as explanatory variables
  - Get predicted response probabilities
  - Use the reciprocal as the weight

- Potential advantages over adjustment cell approach:
  - Can allow a larger number of unit characteristics

2. Propensity Score weighting

- Basic procedure:
  - Define response indicator (*r* =1 if sample unit responds, =0 if not)
  - Fit a regression model with *r* as outcome variable and unit characteristics as explanatory variables
  - Get predicted response probabilities
  - Use the reciprocal as the weight

- Potential advantages over adjustment cell approach:
  - Can allow a larger number of unit characteristics
  - Doesn't have to include all interaction terms (which adjustment cells imply)
  - Include continuous variables

Can also do propensity stratification : possibly more common

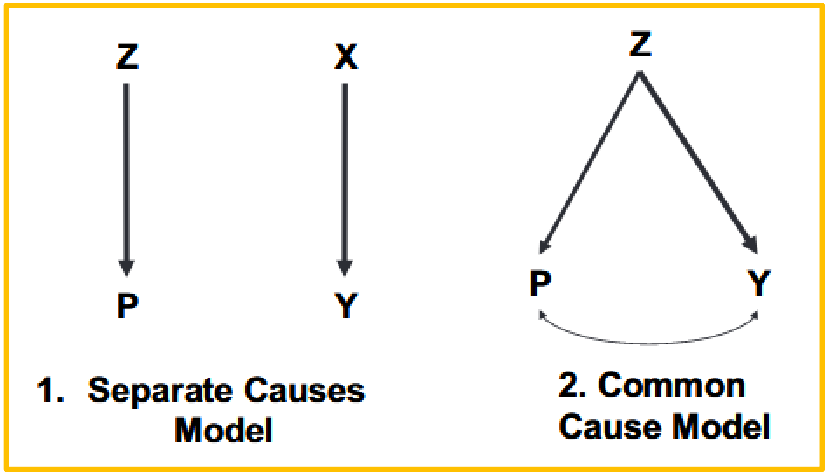
Source of auxiliary variables for nonresponse weighting

- Only characteristics available for all units in the sample may be used

- Alternative sources
  - Enriched sample frame information
  - Paradata

"I just finished fielding a survey. The response rates for seniors and young people are 70% and 15%, respectively. I think I have a big issue with non-response bias."

Which adjustment variables should I choose? The choice has important implications.



Which adjustment variables should I choose? The choice has important implications.

Association with nonresponse	Association with outcome: Low	Association with outcome: High
Low	Cell 1 Bias: --- Var: ---	Cell 2 Bias: --- Var: $\downarrow$
High	Cell 3 Bias: --- Var: $\uparrow$	Cell 4 Bias: $\downarrow$ Var: $\downarrow$

Little and Vartivarian (2005)

Choice of Adjustment Variables…

- Most important features of adjustment variables:
  1. Predictive of outcome of interest
  2. Predictive of nonresponse
- If true, MSE of estimate may be reduced (because both variance and bias will be reduced)
- Important caveat: weighting is done once for entire survey (“global weights”); so impacts vary across individual survey variables

- Yao, Kehinde, Ted, Xiaoqing: How reliable are interviewer observations? What if the auxiliary variables have error ?

Interviewers don't do too badly (West, 2013)

Interviewer judgement: children age < 15 years	*Household roster indicator: children age < 15 years*		*Total*
	No	*Yes*	
No	34898 (59.94%)	9028 (15.51%)	43926 (75.44%)
Yes	7103 (12.20%)	7196 (12.36%)	14299 (24.56%)
Total	42001 (72.14%)	16224 (27.86%)	58225 (100.00%)

Interviewer judgement: selected R sexually active	*Main NSFG interview: selected R sexually active*		*Total*
	No	*Yes*	
No	2230 (9.84%)	2081 (9.18%)	4311 (19.02%)
Yes	2912 (12.85%)	15446 (68.14%)	18358 (80.98%)
Total	5142 (22.68%)	17527 (77.32%)	22669 (100.00%)

But …

Table 4. Adjusted estimates of regression parameters for the two interviewer judgements as predictors of five key NSFG variables, contrasted with estimates by using the true auxiliary variables as predictors instead (NSFG respondents only)†

NSFG variable	*n*	*Pseudo-R^2-values*	*Interviewer judgement: children under 15 years*	*Household roster: children under 15 years*	*Interviewer judgement: sexual activity*	*Respondent report: sexual activity*
Ever been married	22682	0.523/0.571	0.20‡	0.80‡	1.17‡	1.95‡
Ever cohabitated	22682	0.283/0.364	0.13‡	0.32‡	0.98‡	1.99‡
Number of biological children (males only)	10403	0.369/0.466	0.20‡	0.48‡	0.22‡	0.43‡
Number of sexual partners in past year	21008	0.054/0.605	−0.01	−0.05‡	0.26‡	1.17‡
Parity (number of live births) (females only)	12279	0.418/0.489	0.22‡	0.73‡	0.23‡	0.32‡

- Error-prone observations can reduce effectiveness of bias reduction.
- You can also introduce more bias: when $\bar{y}_R > \bar{y}_{NR}$ within each class, and the class level means and response rates are negatively correlated, adjusted estimates will have more bias than complete-case estimates

- Namit:
  - How one can ever know for sure if nonrespondents really differ on key outcomes ?
  - It strikes me how limited our toolkit really is for nonresponse adjustments since most of the time we're stuck with geographic indicators or broad demographics.

- Joy: I'm curious as to whether the lack of variables is primarily due to issues with frame construction, privacy concerns, or the fact that we simply can’t get information from non-responding cases.

- Lingchen: Given the promise of these rarely-used alternative auxiliary variables….what specific theoretical or logistical challenges have historically prevented survey methodologists from routinely incorporating them (paradata) into weighting adjustments ?

Table 6. Main-interview, nonresponse-propensity model predictors: NSFG

Predictor name	Predictor description
MED_HOUSE_VAL_TR_ACS_10_14	Median housing unit value for the Census Tract estimated from ACS 2010-2014
BL_HISP_PERC	Percent of population in Census Block Group that is Non-Hispanic Black or Hispanic from Decennial Census 2010.
MANYUNITS	Interviewer observed characteristic of housing units. Categories include single family home vs multi-unit dwelling.
CHILDRENUNDER15	Interviewer Observation about Presence of Children under the age of 15 in Housing Unit (yes/no)
SEXACTIVE	Contact Obs: Respondent Sex Active (yes/no)
SCR_HISP	Screener interview data indicate selected response is Hispanic (yes/no)
SCR_RACE	Screener interview data indicate selected respondent is Black (yes/no)
SCR_AGE_CAT	Screener interview data on age of selected respondent. Divided into four categories (1= 15-19; 2= 20-29; 3= 30-39; 4= 40-49)
SCR_SEX	Screener interviewer data indicated selected respondent is female (yes/no).
SCR_SINGLEHH	Screener interviewer data indicated selected respondent is only adult in the household (yes/no).
WITHIN_HHPROB	The within-household selection probability of the selected respondent.
MSG_DATA_AVAILABLE	Indicates whether commercial data were successfully merged to the housing unit (yes/no).

https://www.cdc.gov/nchs/data/nsfg/puf3-nsfg-2015-2017-weighting-design_02oct2019.pdf

- Feiran, Angelina: Is it still worthwhile to weight by low-correlation variables?

- Ruisi, Johnia: how should researchers judge whether an auxiliary variable is “sufficiently” correlated, and what concrete statistical thresholds or criteria should be used to decide its inclusion in a weighting model?

- Yujing: “can researchers anticipate in advance which types of paradata are likely to be related to the outcome variables, or is this something that can only be identified through experience or post-hoc analysis”.

Impact of weighting adjustments

- Weighting accounts for *observed* differences between respondents and non-respondents (nonresponse reweighting) or respondents and target population (calibration)
- Does not necessarily account for *unobserved* differences
- Example: 2016 state-level election polls (Kennedy et al. 2017)
  - Standard nonresponse reweighting based on gender and age
  - But education also predictive of vote intention and of the propensity to respond
  - Several state-level estimates of support for Clinton were biased upward

But what was so specific about this election ?

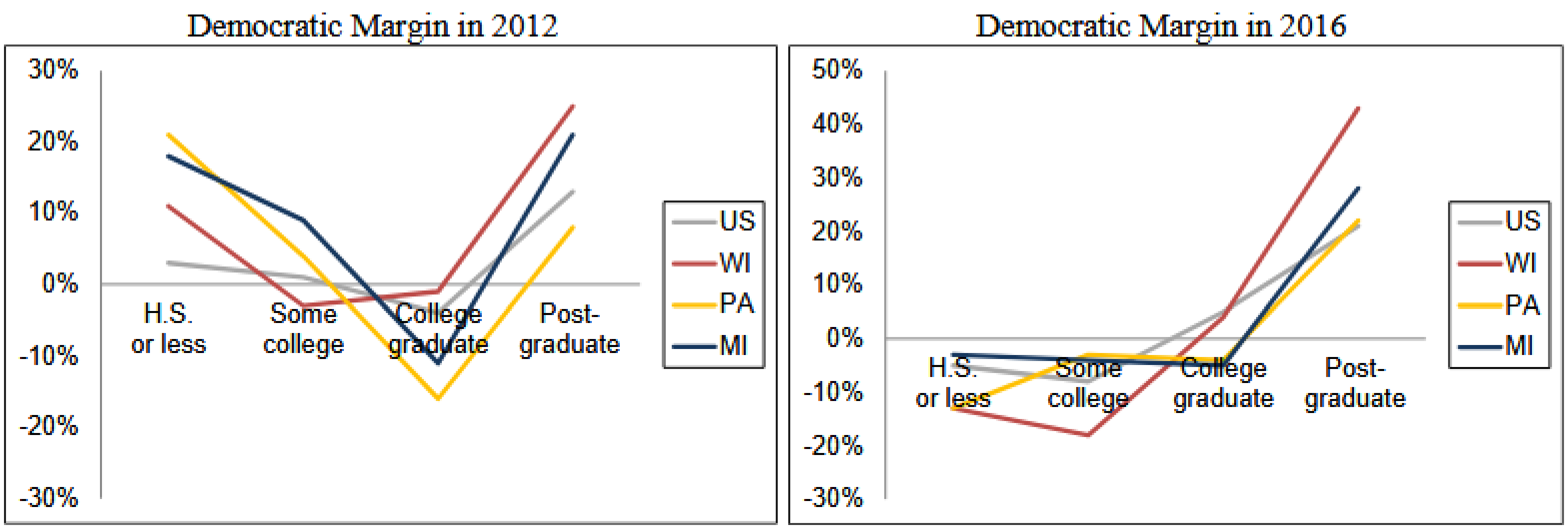


Figure 9. Democratic Presidential Vote Margin in 2012 and 2016 by Voter Education Level and Geography. Source: NEP national Exit Poll 2012, 2016

Sourced from: Kennedy et al 2017
https://aapor.org/wp-content/uploads/2022/11/AAPOR-2016-Election-Polling-Report.pdf

"A number of studies have shown that in general, **people with higher levels of formal education are more likely to take surveys** – it's a very robust finding. Places like Pew Research Center and others have known that for years, and we address that with our statistical weighting – that is, we ask people what their education level is and align our survey data so that it matches the U.S. population on education. And I think **a lot of us assumed that was common practice in the industry** – that roughly speaking, everybody was doing it. And that's not what we found. At the state level, **more often than not, the polls were not being adjusted for education.**

https://www.pewresearch.org/fact-tank/2017/05/04/qa-political-polls-and-the-2016-election/

- **Adjusting for over-representation of college graduates was critical, but many polls did not do it.** In 2016 there was a strong correlation between education and presidential vote in key states. Voters with higher education levels were more likely to support Clinton. Furthermore, recent studies are clear that people with more formal education are significantly more likely to participate in surveys than those with less education. Many polls – especially at the state level – did not adjust their weights to correct for the over-representation of college graduates in their surveys, and the result was over-estimation of support for Clinton.

Weighting Methods

A. Nonresponse (or noncoverage) weighting
- Process of weighting up respondents to account for nonrespondents in sample
- Requires unit-level information on characteristics of both respondents and non-respondents

B. **Post-stratification and Calibration**
- Can improve accuracy of estimates
- Does not require unit level information on characteristics of non-respondents/non-covered units
- Process of adjusting sample to known control totals from some reliable external source

Post-stratification

- Basic procedure:
  - Weights are adjusted so that the weighted totals within mutually exclusive cells equal the known population totals
  - Identify population totals (from external source) for each stratum (N_h)
  - Use the ratio of the two totals ($\frac{N_h}{n_h}$) as the post-stratification adjustment factor
- Example:

Age	Survey Total	Control Total	PS Weight
Age 15-24	10,000	11,000	1.1
Age 25-54	9,000	9,000	
Age 55 plus	11,500	11,000	

Calibration: Raking

- Process of adjusting sample to known control totals from some reliable external source
- Useful when only marginal distributions of control totals are known
  - Iterative process: adjust row totals, then column totals, then row totals and so on until convergence achieved
  - Weighting adjustment factor equals ratio of adjusted cell values to original cell values

Excel example

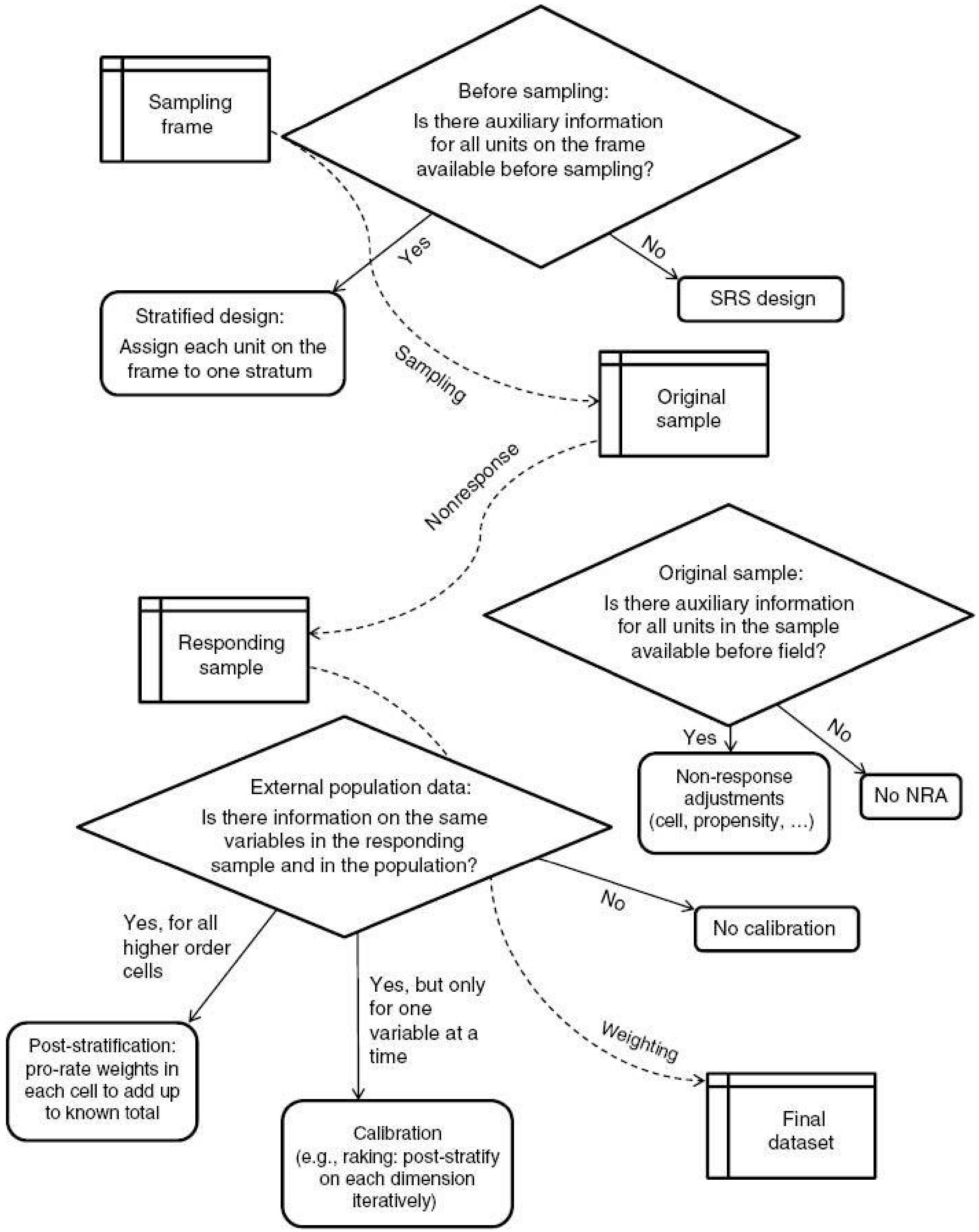
Variables for Calibration

- Any characteristic available for responding units in the sample may be used.
- Demographic variables often used in practice
  - For example: household surveys are commonly calibrated to match characteristics of population by age, sex, race, education, presence/absence of children in household
  - Control totals from external sources are available
  - Way of “keeping up appearances” of survey results

- *https://www.surveypractice.org/article/2809-post-stratification-or-non-response-adjustment* : Clarifies nonresponse adjustment, post-stratification, and calibration.

- Standard post-stratification adjustments may not be entirely effective at removing nonresponse bias from all survey estimates.
  - Some subgroup analyses may be especially subject to bias when adjusting survey estimates using post-stratification alone

(Kreuger and West 2014)



Source: https://www.surveypractice.org/article/2809-post-stratification-or-non-response-adjustment?attachment_id=8953

Adjustment Error

- Error that arises because of errors in post-survey weighting

Potential Sources of Variance

- No association between adjustment variables and survey outcome
- Extreme weights
  - Sometimes caused by small number of respondents in an adjustment cell
  - Increases in the variability of weights increases variance of estimate, other things being equal
  - To address this, units may be divided into larger weighting groups (e.g., quintiles) based on response propensities and/or extreme weights may be trimmed

Potential Sources of Bias

- Impact of measurement errors in paradata?
- Auxiliary variables for nonresponse adjustments are deficient
  - Example: interviewer judgments that are unreliable

"The collection of paradata may not be worthwhile if the resulting data are of reduced quality. Error-prone paradata could lead to biased nonresponse adjustments" (Biemer et al. 2011; West 2011a)

https://www.surveypractice.org/article/3036-paradata-in-survey-research

- West, B. T., & Kreuter, F. (2018). Strategies for increasing the accuracy of interviewer observations of respondent features: Evidence from the US National Survey of Family Growth. *Methodology: European Journal of Research Methods for the Behavioral and Social Sciences, 14*(1), 16–29.

- Asked NSFG interviewers to record open-ended justifications
  1. “He works and goes to school and lives here with his twin – I do not think he could have someone over as the carpet is all taken up and it smells badly of dog poo.”

  2. “He has a tattoo, ‘Carol’, over his heart.”

- Judging based primarily on age will result in systematic false positives. Considering a diversity of cues, including appearance, personality, and other external features improves accuracy.

Potential Sources of Bias

- Auxiliary variables for nonresponse adjustments are deficient
- Control totals for calibration are deficient
- Mismatch between measurement of unit characteristics in survey and in external source for calibration
  - "serious biases or inconsistencies may arise when characteristics used to classify the population and the respondents to the survey are not measured in the same way." (Brick and Kalton 1996)
  - Survey uses different education categories than the American Community Survey
- Clerical errors
  - E.g., error in code used to compute weighting adjustment factors

- In practice, judgement is often involved

  https://www.nytimes.com/interactive/2016/09/20/upshot/the-error-the-polling-world-rarely-talks-about.html

TheUpshot

How Four Pollsters, and The Upshot, Interpreted 867 Poll Responses

We gave four good pollsters the same raw data behind a recent New York Times Upshot/Siena College poll in Florida. They had four different results. Below, their estimates, along with The Upshot's.

				
Clinton +3	**Clinton +1**	**Clinton +4**	**Trump +1**	**Clinton +1**
Charles Franklin Marquette Law	Patrick Ruffini Echelon Insights	Margie Omero, Robert Green, Adam Rosenblatt Penn Schoen Berland Research	Sam Corbett-Davies, Andrew Gelman, David Rothschild Stanford University, Columbia University and Microsoft Research	New York Times Upshot/ Siena College
ESTIMATED ELECTORATE:				
68% white 15% Hispanic 10% black	67% white 14% Hispanic 12% black	65% white 15% Hispanic 12% black	70% white 13% Hispanic 14% black	69% white 14% Hispanic 12% black
LIKELY VOTERS DETERMINED USING:				
Self-report	Vote history	Self-report	Vote history	Self-report and vote history
WEIGHTED TO MATCH:				
Census	Voter File	Voter File	Voter File	Voter File

Using Same Data, 4 Pollsters Arrive at 4 Different Results

- In practice, judgement is often involved

  https://www.nytimes.com/interactive/2016/09/20/upshot/the-error-the-polling-world-rarely-talks-about.html

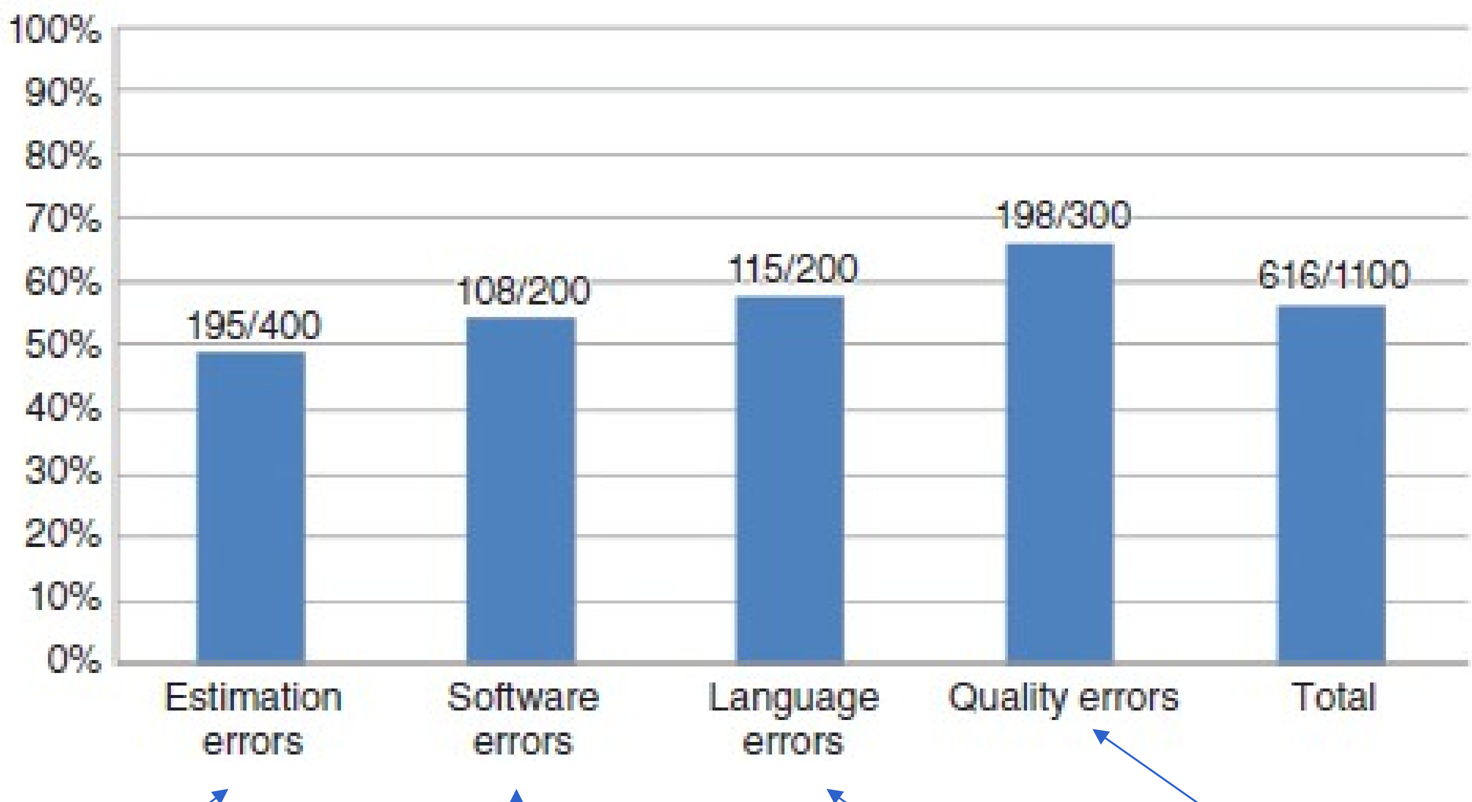
- "Choice of variables for weighting is more consequential than the statistical method."

https://www.pewresearch.org/methods/2018/01/26/how-different-weighting-methods-work/

II. Analytic error

- Under-appreciated component of TSE
- Analytic error: The failure of the survey data user to employ appropriate estimation methods when analyzing the collected survey data
- Examples of indicators: appropriate use of weights for estimation, appropriate use of sampling error codes for variance estimation, use of appropriate software for survey data analysis, appropriate subpopulation analysis approaches, and use of appropriate language to describe the results

Analytic error (West et al 2017)



e.g., weights incorporated?

e.g., PROC SURVEYMEANS mentioned ?

e.g., use “sample estimate” ?

e.g., no mention of the response rate

III. Imputation

Imputation

- Imputation is the placement of one more estimated answers into a field of a data record that previously had no data or had incorrect or implausible data
  - Generally, for item nonresponse in household surveys
  - Plausible data

Levels of Missing Data

- Can be high for certain variables such as income
- Examples:
  - NHIS: roughly 30% of families fail to report an exact value of family income
  - Medicare Current Beneficiary Survey (Marker et al., 2002)
    - Collects cost and payments for each medical event during the past year)
    - 34% complete data; 15% miss both cost and payment data

Common Imputation Methods

- Do nothing
  - Listwise deletion: delete any case with missing values on any variable
  - Pairwise deletion: delete any case with missing values on analytic variables
- Deductive imputation, e.g., marital status for a 12 year boy = “unmarried”
- Use administrative information
- Use prior wave’s value for sample unit, either as is or adjusted to reflect expected change
- ‘Similar’ variables e.g. Impute ‘Total salary’ from individual job salaries

Common Imputation Methods Cont'd

- Mean value imputation
  - Replace missing values for a particular variable with the mean of the item respondent values (sometimes with random noise added)
  - Can be done by subgroup. Example: income imputed by education group
- Hot deck imputation
  - Match "recipients" to "donors".
  - e.g., Missing income. Sort data first by gender and then by education. Move down the list. If a value of income is missing, replace with the value from the most recent unit
- Model-based imputation
  - Predict missing value based on multivariate model using observed data

Multiple Imputation

- Imputation method whereby each missing value is replaced with a vector of possible values
- Basic procedure:
  - Suppose 2 variables: X and Y where you have all info on X
  - Use relationship (a model) between X and Y to impute missing values for Y
  - Create multiple datasets (e.g., 5)
  - Average imputed values across datasets to obtain a single value
  - Associated total variance -> between-variance + within-variance
- Why do this? Most statistical software treats imputed values as real values, leading to estimates of standard errors that are too narrow.
- Multiple imputation allows for estimation of uncertainty about the imputed values ("between-imputation variance")

Implications on Survey Error

- Reduces bias if the missing values can be accurately predicted based on the observed data
  - Possible if adjustment variables are predictive of outcome of interest
- Variance may be reduced because of increase in sample size
  - Example: Imputation of income in NHIS reduces estimated standard errors of regression estimates by 3-25%. Added variance from uncertainty about the imputed values was modest by comparison (Schenker et al., 2006).

Imputation males a difference…

Table 3-11. HLM analysis predicting the narrative performance scores for grade 4 students: Results using the HD imputation and CC analysis

Predictor variable	Hot-deck imputation		Complete cases	
	gamma	standard error	gamma	standard error
Base coefficient	556.1	2.0	564.0	2.0
Proportion minority in class[1]	-64.5	6.7	-64.8	7.0
Gender (1 = female)	14.6	2.3	q	2.7
Age	-1.9	0.2	-1.8	0.2
Minority status[2] (1 = black or Hispanic)	-16.0	3.9	-16.6	4.4
Father education - high school	5.9	4.7	7.7	5.8
Father education - some college	10.0	5.0	12.8	6.1
Father education - college	16.7	4.6	22.5	5.7
No father in household	21.1	7.7	32.6	10.6
Mother education - high school	13.2	4.8	14.0	6.1
Mother education - some college	13.3	5.0	13.3	6.4
Mother education - college	13.8	4.8	12.1	6.2
Family wealth index	6.0	1.3	5.4	1.5
Family composition	16.0	2.4	15.9	2.9
Extended family	-20.2	2.5	-22.8	3.0
Use of foreign language at home	-9.1	2.8	-12.2	3.3
Sample size		6,220		4,280

Methodological Issues in Comparative Educational Studies: The Case of the IEA Reading Literacy Study (1995)

Next Week

- No lecture next week
- Midterm is Monday, October 20th
- I really do not want you guys to use AI. How do we do this ?
- Past years :
  - Take-home, open book, open notes (but **NO AI**)
  - Administered as quiz in Canvas (look under "Quizzes")
  - Most questions will be open-ended questions requiring a short response
  - Can complete between 7:00AM-11:59PM (start by 09:59PM)
  - Two-hour limit once you start

All the best!

(and do reach out for anything)